

Flexible Bayesian Longitudinal Factor Model for Microbiome Counts

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Abstract

Longitudinal microbiome studies provide important insights into the temporal evolution of microbial abundances and dynamic cross feature dependence. Statistical analysis of longitudinal microbiome data is challenging because microbiome counts are highly heterogeneous and contain many zeros, while repeated measurements over time exhibit within-subject temporal dependence and time-varying cross-feature covariance. To address these challenges, we develop a flexible Bayesian longitudinal factor model using a rounded log-normal latent representation. The mean abundance process begins with a Dirichlet process mixture baseline which accommodates sparse and multimodal abundance distributions, and subsequently evolves through a group specific autoregressive process. A separate autoregressive factor process captures shared cross-feature dependence. For identifiability, identity factor innovation covariance fixes the factor scale, while a lower-triangular loading matrix with positive diagonal entries identifies rotations, permutations, and signs. The model therefore distinguishes subject-feature mean evolution from shared dependence and yields a closed-form, time-indexed covariance. Simulation studies evaluate covariance recovery under matched and misspecified count generating mechanisms, and an analysis of longitudinal mouse gut microbiome counts illustrates the identifiable loading structure, distinct temporal processes, and dynamic covariance.

Keywords: Dynamic covariance, Longitudinal microbiome, Bayesian factor model, Dirichlet process mixture

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1 Introduction

High-throughput sequencing (HTS) technologies in microbial ecology generate high-dimensional multivariate counts that characterize communities such as the gut, skin, and oral microbiome. After bioinformatic preprocessing, the data are typically represented as a table of operational taxonomic units (OTUs) or amplicon sequence variants (ASVs) measured across biological samples (Li, 2015). Increasingly, these samples are collected repeatedly from the same subjects to study microbial development, resilience, disease progression, and responses to interventions (Gerber, 2015; Lyu et al., 2023; Creus-Martí et al., 2025). Besides describing how individual taxa change, an important inferential objective is to determine how cross-feature dependence evolves over time.

General microbiome count data present several well-known statistical difficulties. Sequencing depth varies substantially across samples, excess zeros arise from both biological absence and limited detection, nonzero counts are highly overdispersed, and the number of taxa may be comparable to or larger than the number of subjects (Paulson et al., 2013; Li, 2015). Longitudinal sampling introduces two additional challenges. First, repeated samples from the same subject are temporally dependent, violating the independence assumption used by many cross-sectional procedures. Second, temporal variation contains at least two conceptually different components: a subject-feature mean trajectory describing how an individual taxon’s abundance changes, and a shared cross-feature process describing covariance variation among taxa. Without an explicit separation, feature-specific mean evolution can be absorbed into the estimated covariance and create misleading dependence patterns.

Mixture models address an important aspect of microbiome heterogeneity by representing samples or latent parameters as arising from multiple subpopulations. A prominent example is the Dirichlet-multinomial mixture (DMM) (Holmes et al., 2012), which models each microbial composition using one of a finite number of Dirichlet-multinomial components. It accommodates unequal library sizes and overdispersion while clustering samples

into latent metacommunity types. [Gibbons et al. \(2017\)](#) further extended DMM to a longitudinal subject-specific version applied to human gut microbiome time series data. Beyond finite-mixture formulations, Bayesian nonparametric priors have been used in microbiome analysis for several related but distinct purposes. [Ren et al. \(2017\)](#) constructed dependent normalized random measures with latent factors to propagate uncertainty in ordination of microbial communities. [Shuler et al. \(2021\)](#) used dependent DP priors for covariate-indexed distributions of zero-inflation probabilities and differential-abundance effects in multivariate count regression. More recently, Sp-BGFM ([Zhang et al., 2025](#)) modeled multi-domain microbiome count vectors through a DP mixture of rounded multivariate log-normal kernels, coupling flexible baseline abundance distributions with a group factor structure for within- and cross-domain covariance. This formulation demonstrates how DP mixtures can accommodate zeros and substantial between-sample heterogeneity while avoiding an overly restrictive parametric baseline. All these methods use Bayesian nonparametric mixtures for cross-sectional distributions, covariate-indexed effects, or community ordination.

Existing literature addresses high-dimensional microbial dependence in the following two ways. Individuals estimate a sparse covariance or precision matrix after transforming the counts. SparCC ([Friedman and Alm, 2012](#)) estimates log-ratio correlations after sample-total normalization, whereas CCLasso ([Fang et al., 2015](#)) combines a compositional constraint with ℓ_1 regularization. SPIEC-EASI ([Kurtz et al., 2015](#)) applies a centered log-ratio transformation followed by sparse precision-matrix estimation, and Zi-LN ([Prost et al., 2021](#)) extends log-normal graphical modeling to zero-inflated data. Related approaches also include the basis-covariance estimators REBACCA ([Ban et al., 2015](#)) and COAT ([Cao et al., 2019](#)), the compositional graphical lasso of [Tian et al. \(2023\)](#), and the phylogenetically informed PhyloBCG model ([Chung et al., 2022](#)). The second way is to assume a latent structure of the covariance matrix using a small sample size compared to the number of microbes. Latent factor models offer a good example through a low-

rank structure. Dealing with count data directly, ZI-MLN (Zhang et al., 2023) combines a rounded multivariate log-normal kernel with shrinkage priors on the loadings, and Sp-BGFM (Zhang et al., 2025) extends factor modeling to multiple data domains through a joint-sparsity prior. Related low-rank and covariance methods have also been applied to transformed compositional and multi-omics profiles, including MOFA (Argelaguet et al., 2018), robust covariance estimation for compositional data (He et al., 2021), and zero-inflated logistic-normal principal components analysis (Zeng et al., 2023). These methods provide useful cross-sectional estimates, but for longitudinal microbiome data, fitting them independently at each time point discards temporal borrowing, whereas pooling all time points together ignores changes in covariance over time.

Methods developed specifically for longitudinal data have largely emphasized abundance trajectories. MC-TIMME uses negative-binomial observation models with piecewise exponential mean dynamics and clusters OTUs by their responses through a Dirichlet-process mixture (Gerber et al., 2012). Multivariate sparse functional principal-component analysis instead represents subject-level multi-omics trajectories through smooth basis expansions (Jiang et al., 2022). These approaches characterize temporal changes in individual features but are not designed primarily for time-varying covariance estimation. Related dynamic count models include hierarchical negative-binomial state-space models (Chapados, 2014), while dynamic count factor models include gamma-process dynamic Poisson factor analysis (Acharya et al., 2015). Although both approaches borrow information over time, their parametric count distributions may require additional structure to accommodate the zeros and heterogeneous abundance distributions common in microbiome data. MEFISTO introduces Gaussian-process factors for continuous multi-omics measurements (Velten et al., 2022), but analysis of zero-rich microbial counts generally requires a transformation or a separate observation model. Thus, an important gap remains for a model that jointly provides a flexible count distribution, a distinct longitudinal mean process, and an identifiable

factor process for time-varying covariance.

We address this gap with LODI, a Bayesian longitudinal factor model for microbiome counts that explicitly decomposes the temporal signal into a subject-feature mean process and a shared cross-feature factor process. We first use a rounded multivariate log-normal representation from (Canale and Dunson, 2011; Zhang et al., 2025) to handle the discreteness of microbiome counts. Then, we let the initial mean follow a Dirichlet-process mixture (Sethuraman, 1994), allowing heterogeneous and multimodal baseline abundances, and subsequently evolve through a group-specific AR(1) process. Next, we use a time-invariant loading matrix paired with the AR(1) factor process resulting in a dynamic covariance. The abundance and factor components are distinguished through their different hierarchical and temporal structures. We fix the factor scale through identity initial and innovation covariance matrices, while a lower-triangular loading matrix with positive diagonal entries identifies rotations, permutations, and signs (Geweke and Zhou, 1996; Lopes and West, 2004). Different from cross-sectional factor methods such as Sp-BGFM (Zhang et al., 2025), LODI models dependence as a longitudinal process and yields directly identified factor loadings rather than loadings defined only up to an orthogonal transformation. Beyond identified covariance estimation, LODI provides subject and feature specific abundance trajectories whose autocorrelation varies across groups (such as genotypes). In parallel, the shared factor process yields within-time and lagged cross-time covariance matrices, allowing feature-pair dependence to be tracked over time.

The remainder of the paper is organized as follows. Section 2 presents the model, prior specification, identifiability constraints, and posterior computation. Section 3 evaluates LODI in simulation studies, and Section 4 analyzes longitudinal mouse-gut microbiome data. Section 5 concludes with directions for future work.

2 Model

2.1 Rounded log-normal observation model

Let $Y_{ij}(t)$ denote the observed count for feature $j = 1, \dots, J$ from subject $i = 1, \dots, N$ at time $t = 1, \dots, T$. Each subject belongs to a known group $g(i) \in \{1, \dots, G\}$, such as a genotype. Following the rounded-kernel construction of [Zhang et al. \(2025\)](#), we introduce a positive latent variable $Y_{ij}^*(t)$ and its logarithm $\tilde{Y}_{ij}^*(t) = \log Y_{ij}^*(t)$. The latent Gaussian observation model is

$$Y_{ij}^*(t) \sim \text{log-N}(\mu_{ij}(t), \epsilon_{ij}(t)), \quad \tilde{Y}_{ij}^*(t) \sim \text{N}(\mu_{ij}(t), \epsilon_{ij}(t)). \quad (1)$$

Equivalently, $Y_{ij}^*(t)$ is log-normal with log-scale location $\mu_{ij}(t)$ and log-scale variance $\sigma^2(t)$. The observed count is obtained through $Y_{ij}(t) = \lfloor Y_{ij}^*(t) \rfloor$. Then, for every nonnegative integer y ,

$$\begin{aligned} \Pr\{Y_{ij}(t) = y \mid \mu_{ij}(t), \sigma^2(t)\} &= \int_y^{y+1} f_{\text{log-N}}\{y^* \mid \mu_{ij}(t), \sigma^2(t)\} dy^* \\ &= \Phi\{\log(y+1) \mid \mu_{ij}(t), \sigma^2(t)\} - \Phi\{\log y \mid \mu_{ij}(t), \sigma^2(t)\}. \end{aligned} \quad (2)$$

Here, $f_{\text{log-N}}$ denotes the log-normal density, $\Phi(\cdot \mid m, v)$ is the normal distribution function with mean m and variance v . Any order of moments can be calculated from (2) like any discrete distributions.

2.2 Longitudinal mean process

After specifying the rounded kernel, we decompose its complete longitudinal mean as

$$\mu_{ij}(t) = s_i(t) + \pi_{ij}(t) + \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_i(t). \quad (3)$$

The sample-scale term $s_i(t)$ is shared by all features measured from subject i at time t and absorbs multiplicative differences in sequencing depth or total sample abundance. In particular,

$$\mathbb{E}\{Y_{ij}^*(t) \mid s_i(t), \pi_{ij}(t), \boldsymbol{\eta}_i(t)\} = \exp\left\{s_i(t) + \pi_{ij}(t) + \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_i(t) + \frac{\sigma^2(t)}{2}\right\}, \quad (4)$$

so $\exp\{s_i(t)\}$ scales every feature in the same subject-time sample by the same amount. Thus, $s_i(t)$ is a latent sample-scale factor rather than the observed library size itself. Due to the exponential mean formula of log-normal distribution in (4), the conditional likelihood alone does not distinguish the absolute locations of $s_i(t)$ and $\pi_{ij}(t)$. For any scalar d , the transformation $s_i(t) \mapsto s_i(t) + d$, $\pi_{ij}(t) \mapsto \pi_{ij}(t) - d$, leaves $\mu_{ij}(t)$ unchanged. In the hierarchical model, this location invariance is anchored by fixing $s_i(t)$ at the log of the total sum of row counts at each time point $\sum_{j=1} \log(Y_{ij}(t) + 0.01)/J$ or putting a prior with fixed mean 0 which plays a soft constraint as sum to zero constraint in ANOVA model. In the following analysis, we use the second option. The process $\pi_{ij}(t)$ describes subject- and feature-specific abundance after separating the sample-scale and shared-factor contributions. Because it enters the model additively on the log scale, $\pi_{ij}(t)$ corresponds to normalized baseline relative abundance. Equivalently, $\exp\{\pi_{ij}(t) - \pi_{ih}(t)\}$ gives the baseline abundance ratio between features j and h . The final term in (3), $\boldsymbol{\lambda}_j^T \boldsymbol{\eta}_i(t)$, captures lower-dimensional variation shared across features and is responsible for the non-diagonal covariance. We will discuss it in detail in Section 2.4.

2.3 Longitudinal Dirichlet-Process Mixture

Microbial abundances are often sparse, skewed, and multimodal across subjects and features. A common strategy for excess zeros is a zero-inflated model that combines a zero-generating mechanism with a parametric model for abundance. Examples include the zero-inflated Gaussian model in metagenomeSeq for differential-abundance analysis (Paulson

et al., 2013), the zero-inflated log-normal model for microbial association networks (Prost et al., 2021), and the zero-inflated logistic-normal multinomial model for estimating microbial compositions (Zeng et al., 2023). These formulations can distinguish structural absence from sampling variation, but they require a separate zero-generating mechanism and assumptions that identify it from the abundance component. Moreover, an observation attributed to the structural-zero component does not directly inform the positive-abundance density. In contrast, the rounded log-normal kernel in (2) assigns positive probability to zero through interval censoring, so zero and nonzero counts inform a common latent abundance distribution. To accommodate remaining multimodality without imposing a single parametric baseline distribution, we place a Dirichlet-process (DP) mixture prior on the initial value of the log-normalized abundance process:

$$\pi_{ij}(1) \sim \sum_{\ell=1}^{\infty} w_{\ell} \mathcal{N}(\xi_{\ell}, \tau_{\ell}^2), \quad (5)$$

$$\pi_{ij}(t+1) = \phi_{g(i)} \pi_{ij}(t) + v_{ij}(t+1), \quad v_{ij}(t+1) \sim \mathcal{N}(0, V). \quad (6)$$

The mixture in (5) clusters the NJ subject-feature baselines probabilistically while allowing the number of occupied clusters to be learned from the data. The subsequent group-specific AR(1) process propagates each baseline through time. Here, $\phi_g \in (-1, 1)$ is the first-order autocorrelation parameter in AR(1) for group g , and V is the innovation variance. Using the stick-breaking representation (Sethuraman, 1994), we let $w_{\ell} = z_{\ell} \prod_{h < \ell} (1 - z_h)$, $z_{\ell} \stackrel{iid}{\sim} \text{Be}(1, c)$. We use the following conditionally conjugate priors $\xi_{\ell} \stackrel{iid}{\sim} \mathcal{N}(\xi_0, \kappa_0^2)$, $\tau_{\ell}^2 \stackrel{iid}{\sim} \text{inv-Ga}(a_{\tau}, b_{\tau})$, $\phi_g \stackrel{iid}{\sim} \mathcal{N}(\mu_{\phi}, \sigma_{\phi}^2) \mathbb{1}\{-1 < \phi_g < 1\}$, $V \sim \text{inv-Ga}(a_V, b_V)$. Throughout, $\text{inv-Ga}(a, b)$ denotes an inverse-gamma distribution with shape a and rate b .

2.4 Longitudinal factor process

Let $\boldsymbol{\eta}_i(t) \in \mathbb{R}^K$ be the factor process for subject i , let $\boldsymbol{\lambda}_j \in \mathbb{R}^K$ be the loading vector for feature j , and let $\boldsymbol{\Lambda} = (\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_J)^\top$. We let the factors follow AR(1) process

$$\boldsymbol{\eta}_i(1) \sim N(\mathbf{0}, I_K), \quad (7)$$

$$\boldsymbol{\eta}_i(t+1) = \rho \boldsymbol{\eta}_i(t) + \mathbf{u}_i(t+1), \quad \mathbf{u}_i(t+1) \sim N(\mathbf{0}, I_K), \quad (8)$$

with autocorrelation parameter $\rho \sim N(\mu_\rho, \sigma_\rho^2) \mathbb{1}\{-1 < \rho < 1\}$. In a Bayesian factor model, the likelihood depends on the loadings and factors through their product. For any non-singular $K \times K$ matrix H , $\boldsymbol{\Lambda} \boldsymbol{\eta}_i(t) = (\boldsymbol{\Lambda} H) \{H^{-1} \boldsymbol{\eta}_i(t)\}$, so the loading matrix and factor coordinates are not separately identified without additional restrictions. Fixing both the initial and innovation covariance matrices of the factors at I_K fixes their scale and restricts this invariance to orthogonal transformations. To remove the remaining rotation, permutation, and sign ambiguities, we constrain the first K rows of $\boldsymbol{\Lambda}$ to be lower triangular with positive diagonal:

$$\lambda_{jk} = 0 \quad \text{for } k > j, \quad \lambda_{jj} > 0 \quad \text{for } j = 1, \dots, K.$$

Provided that the leading $K \times K$ block has full rank, the triangular restriction fixes the factor orientation and ordering, while the positive diagonal fixes the signs (Geweke and Zhou, 1996; Lopes and West, 2004). We place independent $N(0, 1)$ priors on the free loading entries and let $\sigma^2(t) \stackrel{iid}{\sim} \text{inv-Ga}(a_\sigma, b_\sigma)$ for conjugacy.

Let $\tilde{\mathbf{Y}}_i^\star(t) = \{\tilde{Y}_{i1}^\star(t), \dots, \tilde{Y}_{iJ}^\star(t)\}^\top$, after integrating out the shared factor score, (7)–(8) imply that the within-time covariance of the latent log-abundance vector is

$$\boldsymbol{\Sigma}(t) \equiv \text{Cov}\left\{\tilde{\mathbf{Y}}_i^\star(t) \mid s_i(t), \boldsymbol{\pi}_i(t)\right\} = \frac{1 - \rho^{2t}}{1 - \rho^2} \boldsymbol{\Lambda} \boldsymbol{\Lambda}^\top + \sigma^2(t) I_J. \quad (9)$$

For a positive lag $\delta \geq 1$, the corresponding cross-time covariance is

$$\text{Cov}\left\{\tilde{\mathbf{Y}}_i^*(t), \tilde{\mathbf{Y}}_i^*(t + \delta) \mid s_i(t), \boldsymbol{\pi}_i(t), s_i(t + \delta), \boldsymbol{\pi}_i(t + \delta)\right\} = \rho^\delta \frac{1 - \rho^{2t}}{1 - \rho^2} \mathbf{\Lambda} \mathbf{\Lambda}^\top. \quad (10)$$

These latent covariance matrices also determine whether two observed counts are uncorrelated. In particular, conditional on $s_i(t)$ and $\boldsymbol{\pi}_i(t)$, $\tilde{\mathbf{Y}}_i^*(t)$ is multivariate normal. Thus, for $j \neq j'$, $\Sigma_{jj'}(t) = 0$ implies that $\tilde{Y}_{ij}^*(t)$ and $\tilde{Y}_{ij'}^*(t)$ are independent. Because the count transformation is applied separately to each coordinate, their transformed counts are also independent, and hence

$$\Sigma_{jj'}(t) = 0 \implies \text{Cov}\{Y_{ij}(t), Y_{ij'}(t) \mid s_i(t), \boldsymbol{\pi}_i(t)\} = 0.$$

The same implication applies across time when the corresponding entry of (10) is zero. When a latent covariance is nonzero, the associated counts are generally dependent. We therefore conduct population covariance analyses on the latent log-abundance scale using (9) and (10).

Our formulation uses a common loading matrix $\mathbf{\Lambda}$ across time so that the same identified factor directions describe dependence throughout follow-up. A more general specification could replace it by $\mathbf{\Lambda}(t)$, giving within-time covariance $\{(1 - \rho^{2t})/(1 - \rho^2)\} \mathbf{\Lambda}(t) \mathbf{\Lambda}(t)^\top + \sigma^2(t) I_J$ and cross-time covariance $\rho^\delta \{(1 - \rho^{2t})/(1 - \rho^2)\} \mathbf{\Lambda}(t) \mathbf{\Lambda}(t + \delta)^\top$. Such a model permits pair-specific changes in covariance beyond the common temporal scaling in (9). However, estimating a separate $J \times K$ loading matrix at each time point is difficult when the number of subjects observed per time point is small relative to the number of features, as in microbiome studies with $N \ll J$. The time-invariant loading matrix therefore pools information across time and provides a more stable covariance estimate, but time-varying loadings would require additional temporal smoothing or shrinkage.

2.5 Posterior computation

Let Θ collect all unknown quantities, for an observed count $y_{ij}(t)$ we introduce the augmented latent Gaussian variable restricted to $[\log y_{ij}(t), \log\{y_{ij}(t)+1\})$, with lower endpoint $-\infty$ when $y_{ij}(t) = 0$. The augmented posterior is proportional to

$$\prod_{i=1}^N \prod_{j=1}^J \prod_{t=1}^T \mathcal{N}[\tilde{y}_{ij}^*(t) \mid s_i(t) + \pi_{ij}(t) + \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_i(t), \sigma^2(t)] \\ \times \mathbb{1}[\log y_{ij}(t) \leq \tilde{y}_{ij}^*(t) < \log\{y_{ij}(t) + 1\}] p(\Theta). \quad (11)$$

Both longitudinal trajectories, $\{\pi_{ij}(t)\}_{t=1}^T$ and $\{\boldsymbol{\eta}_i(t)\}_{t=1}^T$, are sampled jointly over time using forward-filtering backward-sampling (FFBS). One MCMC iteration proceeds as follows.

1. Sample each $\tilde{Y}_{ij}^*(t)$ from its normal full conditional truncated to $[\log y_{ij}(t), \log\{y_{ij}(t) + 1\})$, using $\log(0) = -\infty$.
2. Sample the latent sample-scale term $s_i(t)$. Under $s_i(t) \sim \mathcal{N}(m_s, v_s)$,

$$s_i(t) \mid \cdot \sim \mathcal{N}\{m_{s,it}, C_{s,t}\}, \\ C_{s,t} = \left\{ \frac{J}{\sigma^2(t)} + \frac{1}{v_s} \right\}^{-1}, m_{s,it} = C_{s,t} \left[\frac{m_s}{v_s} + \frac{1}{\sigma^2(t)} \sum_{j=1}^J \{\tilde{y}_{ij}^*(t) - \pi_{ij}(t) - \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_i(t)\} \right].$$

3. Sample the trajectory $\{\pi_{ij}(t)\}_{t=1}^T$ by FFBS, and update the mixture atoms, stick-breaking weights, group-specific coefficients ϕ_g , and innovation variance V .
4. Sample the free elements of Λ subject to its triangular and positive-diagonal restrictions. Update each trajectory $\{\boldsymbol{\eta}_i(t)\}_{t=1}^T$ by FFBS, center the factor scores across subjects at every time point, and then update ρ .

5. Independently for each time point, sample

$$\sigma^2(t) \mid \cdot \sim \text{inv-Ga} \left[a_\sigma + \frac{NJ}{2}, b_\sigma + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^J \{ \tilde{y}_{ij}^*(t) - s_i(t) - \pi_{ij}(t) - \boldsymbol{\lambda}_j^\top \boldsymbol{\eta}_i(t) \}^2 \right].$$

Complete full-conditional derivations are provided in the supplementary material. The sampler is implemented in Rcpp and RcppArmadillo for computational efficiency. R package is available on <https://shuang-jie.github.io/LODI>.

3 Simulation

3.1 Setup and evaluation

We evaluate LODI’s performance on a sequence of simulation studies that progressively introduce real-data features. We first test whether LODI identifies factors and recovers the true covariance under matched-model and ZINB-misspecified conditions. Then, we utilize real-data features in Section 4 to generate a simulation that mimics real, challenging data. In all simulations we generate longitudinal count data $\mathbf{Y} \in \mathbb{Z}_{\geq 0}^{N \times T \times J}$ and measure recovery of the covariance $\boldsymbol{\Sigma}(t)$ implied by the truth using $\text{RMSE}(t) = \sqrt{\sum_{j,j'} (\hat{\boldsymbol{\Sigma}}(t)_{jj'} - \boldsymbol{\Sigma}^{\text{true}}(t)_{jj'})^2 / J^2}$. We report RMSE averaged across time within each replicate and then across the independent replicates available for each simulation setting.

We compare LODI to three covariance estimators commonly used in microbiome data analysis: (i) SP-BGFM (per-time) (Zhang et al., 2025), a Sparse Bayesian Group Factor Model fit separately at each timepoint; (ii) SP-BGFM (multi-domain), the same model with all timepoints stacked as separate domains; and (iii) MEFISTO (Velten et al., 2022), a factor model with a Gaussian process on longitudinal covariates. SP-BGFM is fitted directly to the observed counts, whereas MEFISTO uses $\log(Y + 0.5)$ as Gaussian input.

The implementation fixes $(\xi_0, \kappa_0^2, a_\tau, b_\tau, \mu_\phi, \sigma_\phi^2, a_V, b_V, c) = (0, 25, 2, 1, 0, 1, 2, 1, 1)$. Thus,

the mixture-location prior has standard deviation 5, and the DP concentration parameter is fixed at $c = 1$. Our results are not sensitive to these non-informative prior choices. The implemented sampler uses a finite truncation with L mixture components, and the simulation and data analyses use $L = 30$. We examine the sensitivity of LODI to $K \in \{4, 6, 8, 10\}$ when the data-generating truth has $K^{\text{tr}} = 4$ in the simulation. Representative MCMC trace plots and additional parameters and covariance results are provided in Section 2 of the supplementary material.

3.2 Simulation results

Sim 1 generates counts directly from the LODI model to evaluate mean-process and covariance recovery under matched conditions and to assess sensitivity to the fitted number of factors. The data-generating design uses $N = 20$, $T = 7$, $J = 50$, and $K^{\text{tr}} = 4$. The baseline $\pi_{ij}(1)$ has three mixture components with locations $\{-5, 3, 8\}$, common variance 0.5, and weights $\{0.4, 0.3, 0.3\}$. The group-specific abundance-process AR(1) coefficients are $\phi_g^{\text{tr}} \in \{-0.8, 0.9, 0.7\}$ with $V = 0.2$. The factor process has $\rho^{\text{tr}} = 0.7$, initial covariance I_K , and innovation covariance I_K . $\sigma^{2,\text{tr}}(t)$ is drawn independently from $\text{Unif}(0, 1)$. Finally, counts are generated as $Y_{ij}(t) = \lfloor \exp\{\tilde{Y}_{ij}^*(t)\} \rfloor$. We fit LODI using $K \in \{4, 6, 8, 10\}$ and fit each comparator using its standard specification.

Figure 1 illustrates recovery of the two model components that are directly interpretable under LODI’s identifying restrictions. In (a), the posterior mean of $\mu_{ij}(t) = s_i(t) + \pi_{ij}(t)$ closely follows its truth value across subjects, features, and times. The group-specific AR coefficients are also accurately recovered, with posterior means (SDs) of -0.801 (0.004), 0.910 (0.004), and 0.694 (0.006), compared with the true values -0.8 , 0.9 , and 0.7 , respectively. A good estimate of the mean provides a baseline for the covariance estimate. We compare entries of $\mathbf{\Lambda}$ directly with the truth in (b) since both the true and fitted loading matrices obey the same lower-triangular, positive-diagonal constraint. No post-hoc rotation

or sign alignment is used. This also promises a good estimate of covariance in Table 1.

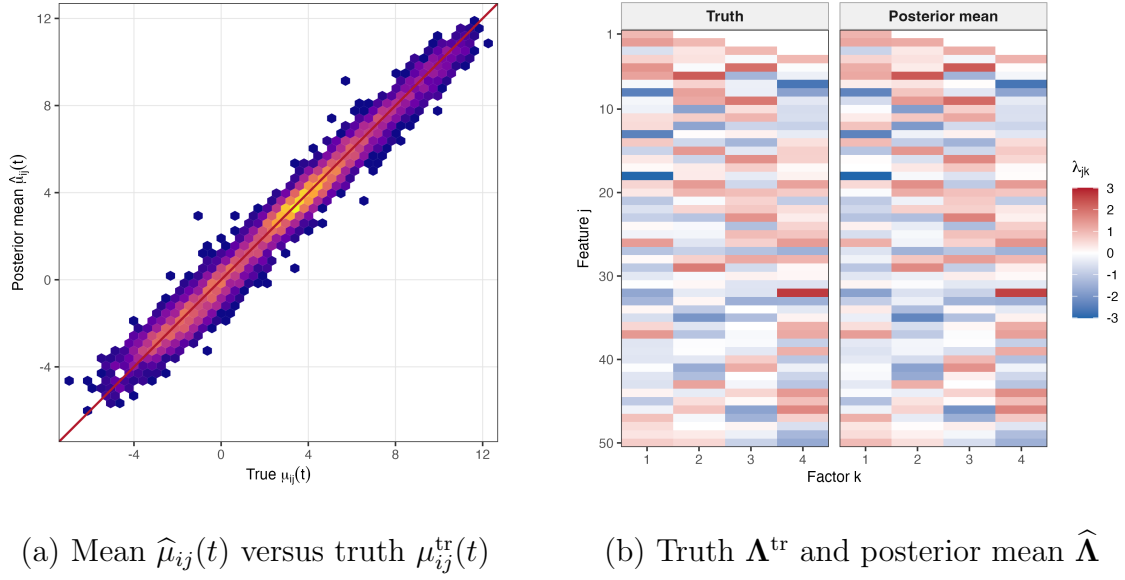


Figure 1: [Simulation 1] (a) Posterior mean versus truth for all $N \times T \times J = 7,000$ values of the longitudinal mean process $\mu_{ij}(t)$; warmer color represents a larger number of overlapping values, and the red line is the identity line. (b) Heatmaps of the true and posterior mean loading matrices with features in rows and factors in columns.

LODI substantially outperforms all three comparators and remains stable under moderate over-specification of the factor dimension in Table 1. The estimates of covariance for $K = 4, 6, 8$, and 10 are nearly indistinguishable relative to their average RMSE. The details of other parameter estimates and different K are listed in the Supplementary. These results indicate that covariance recovery is identifiable and not sensitive to over-specification of K .

Method	Fitted K	Average RMSE (SE)
LODI	4 (truth)	0.840 (0.024)
LODI	6	0.853 (0.020)
LODI	8	0.837 (0.020)
LODI	10	0.883 (0.021)
MEFISTO	–	3.165 (0.036)
SP-BGFM (per-time)	–	3.557 (0.042)
SP-BGFM (multi-domain)	–	3.742 (0.046)

Table 1: [Simulation 1] RMSE of $\Sigma(t)$ covariance recovery is averaged across time within each replicate and then across 50 independent replicates. Parentheses report its standard error.

In Sim 2, we sample the simulated data with a zero-inflated negative binomial distribution which is misspecified from our truth model. Let $z_{ij}(t) \sim \text{Bern}(\pi)$ denote structural absence. If $z_{ij}(t) = 1$, then $Y_{ij}(t) = 0$; otherwise $Y_{ij}(t) \sim \text{NB}(\vartheta_{ij}(t), \phi_{\text{NB}} = 3)$. Since the negative binomial distribution does not have a covariance structure, we put the latent factors on the log of the latent mean. $\log \vartheta_{ij}(t) = c_i + \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_i(t)$ and $c_i \sim \text{N}(3, 1)$. The factor process has $\rho = 0.7$ and identity innovation covariance. Thus the data-generating likelihood is misspecified relative to LODI. We use $N = 20$, $T = 5$, $J = 50$, $K = 4$, and test the zero inflation rate $\pi \in \{0.10, 0.30, 0.50\}$.

Method	$\pi = 0.10$	$\pi = 0.30$	$\pi = 0.50$
LODI	0.954 (0.029)	1.205 (0.036)	1.657 (0.053)
MEFISTO	2.194 (0.068)	2.672 (0.067)	2.974 (0.085)
SP-BGFM (per-time)	2.241 (0.073)	2.849 (0.083)	3.290 (0.085)
SP-BGFM (multi-domain)	2.477 (0.082)	2.926 (0.072)	3.298 (0.092)

Table 2: [Simulation 2] RMSE of $\Sigma(t)$ covariance recovery under increasing zero inflation is averaged across time within each replicate and then across 50 replicates. Parentheses report its standard error.

LODI performs best relative to the comparators across the three values of zero rates π in Table 2. This advantage over SP-BGFM cannot be attributed to count preprocessing, because both methods use rounded-log-normal augmentation for the raw counts. Instead, the comparison highlights LODI’s autoregressive factor process, which shares information across time while directly representing the evolution of $\Sigma(t)$. Fitting SP-BGFM separately at each timepoint discards this temporal borrowing, whereas treating timepoints as multiple domains does not impose an ordered temporal process. MEFISTO models smooth temporal factors but is fitted to $\log(Y + 0.5)$ without a flexible mean process. Additional results for Simulation 2 are provided in the supplementary material.

Sim 3 evaluates real-data scale simulation under the zero-inflated negative-binomial misspecification introduced in Sim 2. In real data Section 4, we have 20 samples collected from 7 time points with 54 OTUs. Then we use $N = 20$, $T = 7$, $J = 54$, and $K = 4$, and generate the latent log mean as $\log \vartheta_{ij}(t) = c_i + \lambda_j^T \boldsymbol{\eta}_i(t)$. To reproduce the count scale, we draw $c_i \sim N(\bar{\ell}, s_\ell^2)$ using the empirical log-library-size mean $\bar{\ell} \approx 10.76$ and empirical standard deviation $s_\ell \approx 0.77$. We set the structural-zero probability to the empirical zero rate, $\pi = 0.189$, and use a factor process with $\rho = 0.7$ and identity innovation covariance.

Figure 2 examines individual longitudinal recovery and posterior predictive checking for goodness of fit. Panel (a) shows four posterior mean estimates of mean trajectories (solid) with distinct temporal shapes compared with the true normalized latent log-mean processes (dashed). For the posterior predictive checking, we then compare summaries of posterior predictive samples with the corresponding summaries of the observed or simulated counts. Panels (b)–(d) provide distributional checks across increasing sparsity. Panel (b) evaluates the positive-count distribution in a setting with no observed zeros, and panels (c) and (d) represent increasingly sparse settings. As the zero rate in the observed data increases, the posterior predictive distribution correspondingly places more mass near zero. This pattern indicates that the rounded kernel reproduces zero-rich count behavior without a separate

zero-inflation indicator.

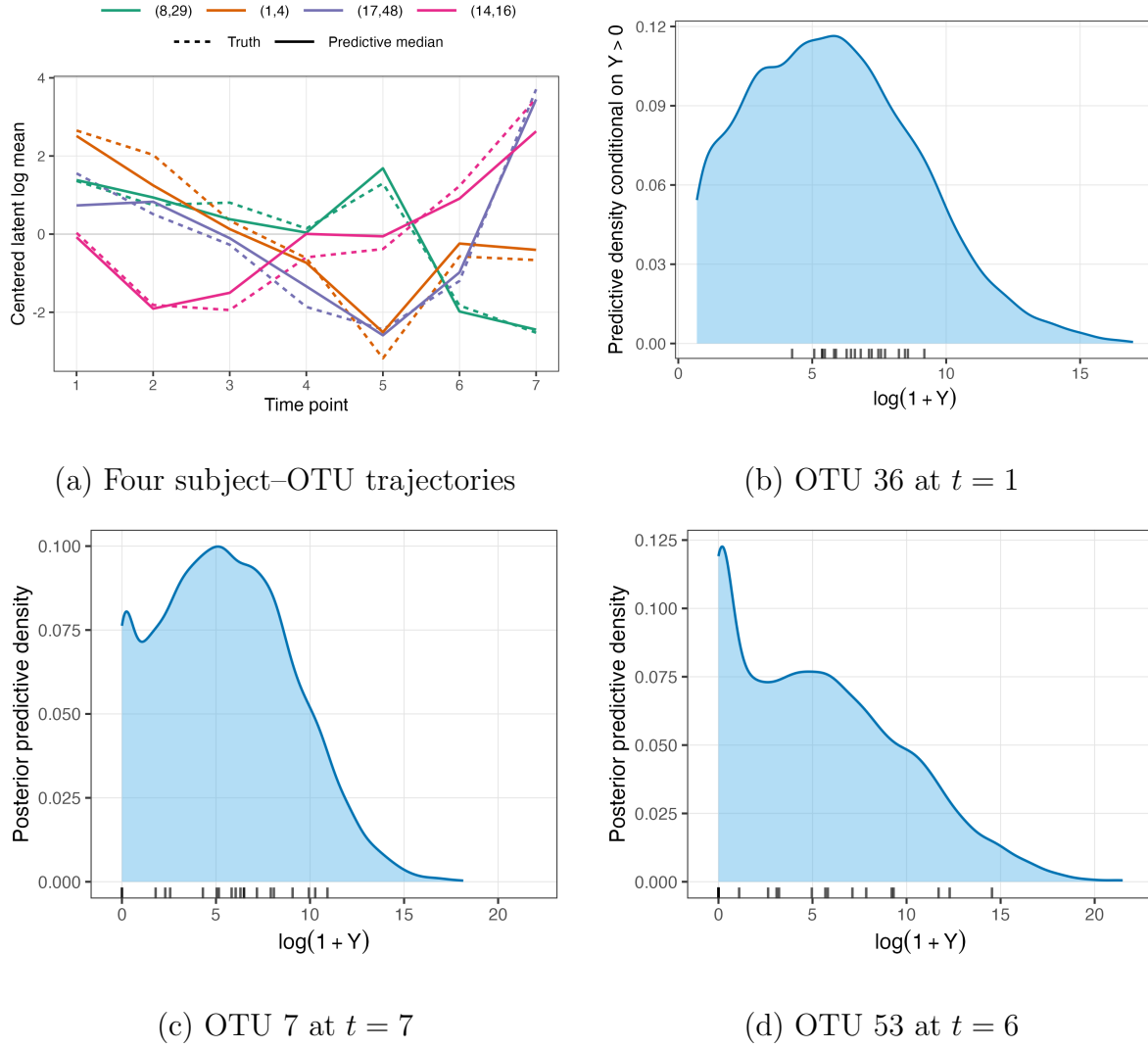


Figure 2: [Simulation 3] Panel (a) compares true normalized latent log-means (dashed) with posterior mean estimates (solid) for $(i, j) = (8, 29), (1, 4), (17, 48),$ and $(14, 16)$. Panels (b)–(d) show posterior predictive densities of $\log(Y_{ij}^{1+\text{rep}})$ for OTU 36 at $t = 1$, OTU 7 at $t = 7$, and OTU 53 at $t = 6$, respectively. Black rug marks denote the 20 observed counts.

Table 3 evaluates covariance recovery for LODI and comparators across all replicates. LODI has the smallest average RMSE. The absolute RMSE values are not directly comparable with those in Simulation 2 because Simulation 3 uses different dimensions and a distinct real-data-calibrated data-generating setting.

Method	Average RMSE (SE)
LODI	2.002 (0.039)
MEFISTO	2.376 (0.057)
SP-BGFM (per-time)	2.284 (0.074)
SP-BGFM (multi-domain)	2.976 (0.083)

Table 3: [Simulation 3] RMSE of $\Sigma(t)$ covariance recovery is averaged across time within each replicate and then across 50 replicates, and parentheses report standard errors.

4 Real data analysis

We apply LODI to a longitudinal mouse-gut microbiome dataset comprising $N = 20$ mice measured at $T = 7$ timepoints across $J = 54$ OTUs. Subjects belong to three genotype groups (ACK/G 4, HACK/G 10, WT 6). The raw count matrix has a total observation count summing to $\approx 1.6 \times 10^7$, with library sizes per (subject, time) ranging 1.0×10^4 to 3.1×10^5 . The overall zero rate is 18.9%, but per-feature zero rates are highly heterogeneous. We fit LODI with $K = 10$ latent factors and 400,000 MCMC iterations.

To assess goodness of fit, we conduct posterior predictive checks and compare the fitted abundance process to the observed one. Figure 3 shows the posterior predictive density of $\log(Y_{ij}^{\text{rep}} + 1)$ for six representative OTUs ordered by their observed zero rates. LODI’s posterior predictive distribution captures the shape of the count distribution flexibly for abundant OTUs with few zeros (e.g., OTU 32, zero rate 0%), bimodal OTUs (OTU 52, zero rate 18%), and sparse OTUs (e.g., OTU 42, zero rate 55%). This flexibility benefits directly from LODI’s DP-mixture prior on $\pi_{ij}(1)$.

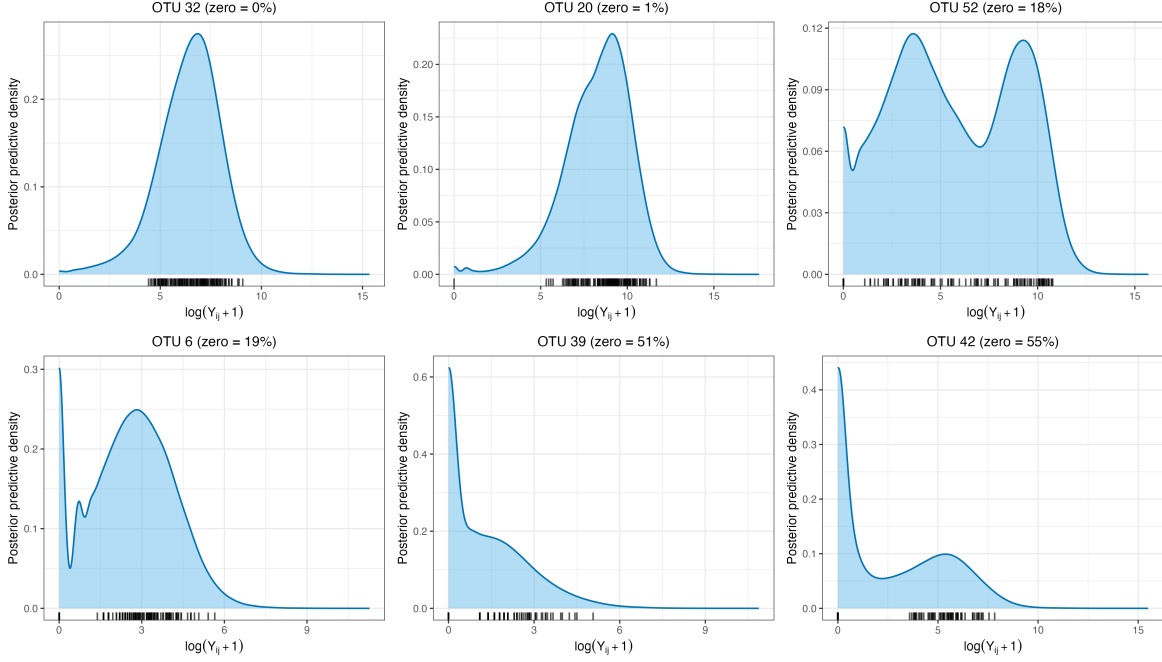


Figure 3: [Real Data] Posterior predictive density (blue) of $\log(Y_{ij}^{\text{rep}} + 1)$ for six representative OTUs ordered by increasing observed zero rate, with observed counts shown as black rug marks. OTU 52 illustrates the bimodal flexibility enabled by the DP-mixture baseline for $\pi_{ij}(1)$, and OTU 6 illustrates recovery under moderate sparsity. Sparse OTUs 39 and 42 show substantial concentration near zero.

Figure 4(a) illustrates the fitted normalized abundance process for four subject–OTU pairs. The empirical curves are smoothed log-relative abundances, and the fitted curves are posterior means of the normalized latent process. Both are centered over time to compare their temporal shapes. The posterior trajectories reproduce several rises, declines, and reversals in the observed data. These posterior checks indicate LODI distinguishes the two underlying mechanisms well. The posterior means of ϕ_g , which capture the feature-specific abundance process $\pi_{ij}(t)$, are 0.993, 0.994, and 0.995 across the three genotype groups, and the innovation variance is $V = 0.473 (\pm 0.040)$. Thus, $\pi_{ij}(t)$ is highly persistent. In contrast, the shared factor process has posterior mean $\rho = 0.347 (\pm 0.065)$ and 95% credible interval $[0.221, 0.474]$, indicating subject-specific changes across OTUs. Figure 4(b) displays the posterior mean of the identified loading matrix. Its effective rank is substantially smaller than the fitted $K = 10$. Measured by $\|\hat{\mathbf{\Lambda}}_{\cdot k}\|^2 / \|\hat{\mathbf{\Lambda}}\|_F^2$, factors 1, 3, and 5 contribute 56.4%,

21.6%, and 8.2%, respectively, whereas each remaining factor contributes only 1.2%–2.7%.

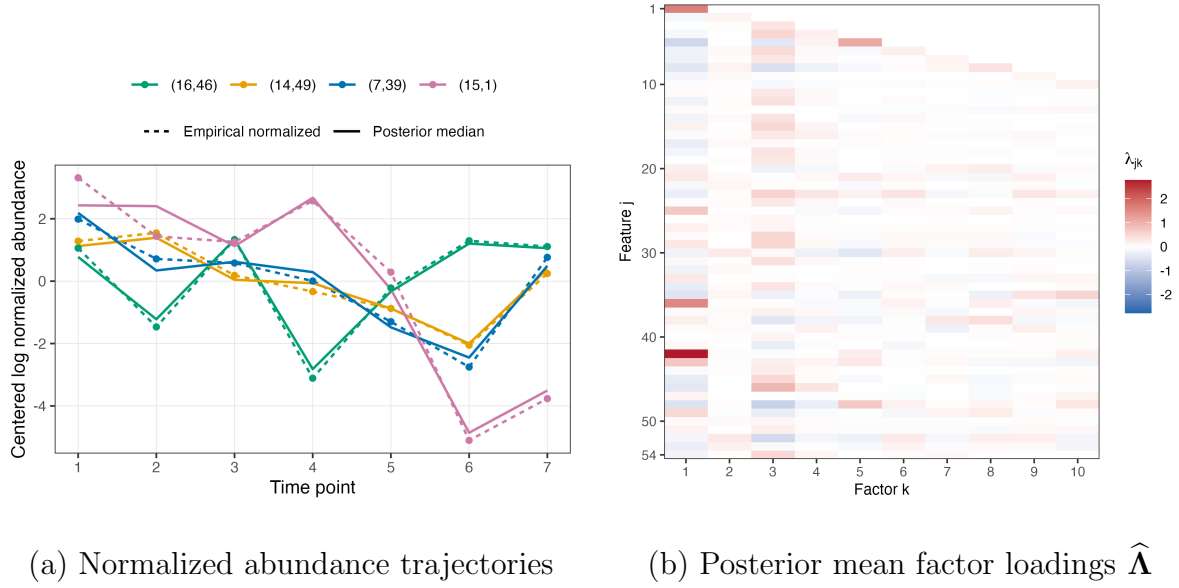


Figure 4: [Real Data] In (a), empirical smoothed log-relative abundances (dashed, with points) are compared with posterior mean normalized latent trajectories (solid) for four illustrative subject–OTU pairs; each trajectory is centered over time. Panel (b) shows the posterior mean of the identified loading matrix $\hat{\Lambda}$.

Motivated by the three dominant loading columns in Figure 4(b), Figure 5 shows posterior mean trajectories of $\eta_{ik}(t)$ for factors 1, 3, and 5. Thin curves represent individual subjects, and thick curves represent genotype-group averages. Factor 1 exhibits the strongest short-term reversals, including an ACK/G decline from W1 to W3 followed by recovery at W7 and a WT decline at W7 followed by recovery at W19. Factor 3 shows increasing separation after W7: the WT group moves upward, reaching 0.71 at W19, whereas HACK/G moves downward to -0.46 . Factor 5 remains closer to zero at the group level, although individual subjects still fluctuate.

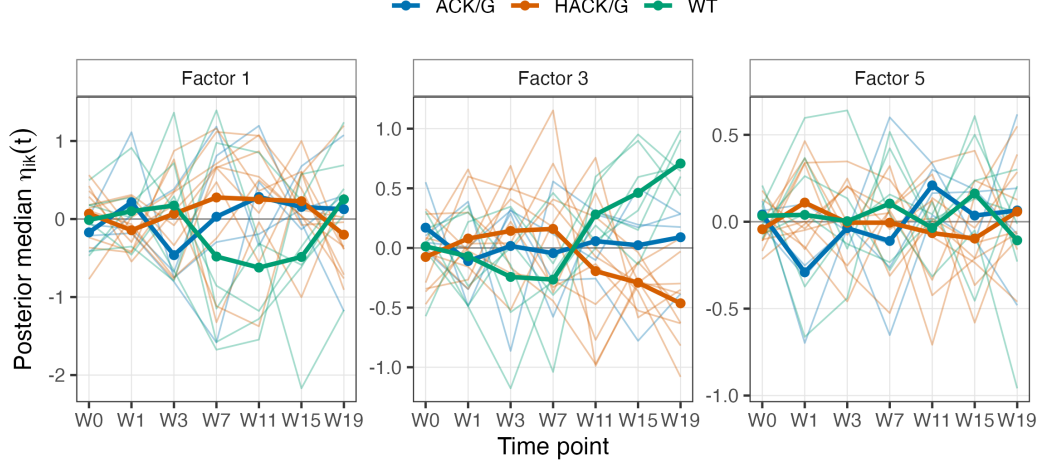


Figure 5: [Real Data] Posterior mean factor-score trajectories $\eta_{ik}(t)$ for the three dominant loading directions. Thin curves show individual subjects and thick curves show genotype-group averages. Factor scores are centered across subjects at each time point.

The magnitude of λ_{jk} measures how strongly OTU j participates in factor k , while its sign determines whether the OTU changes in the same or opposite direction as the factor score. Factor 1 is driven most strongly by OTUs 42 (*Flavobacterium*; posterior mean loading 2.77), 1 (*Clostridium sensu stricto 1*; 1.65), and 36 (*Caldalkalibacillus*; 1.56). The largest absolute loadings for factor 3 occur for OTUs 46 (Chloroplast; 0.93), 48 (*Ileibacterium*; -0.77), and 52 (Erysipelotrichaceae; -0.64). Factor 5 is driven primarily by OTU 5 (Pasteurellaceae; 1.12), followed by OTUs 48 (*Ileibacterium*; 0.71) and 30 (-0.40).

5 Discussion

We have introduced LODI, an identifiable Bayesian longitudinal factor model for microbiome counts. It combines a DP-mixture baseline and group-specific AR(1) abundance process $\pi_{ij}(t)$ with a shared AR(1) factor process $\boldsymbol{\eta}_i(t)$. Identity initial and innovation covariance matrices fix the factor scale, while the lower-triangular positive-diagonal loading constraint identifies rotation, permutation, and sign. Posterior inference uses FFBS updates for both longitudinal processes within an Rcpp-based Gibbs sampler. Simulations

under both the assumed model and misspecified zero-inflated negative-binomial mechanisms demonstrate improved covariance recovery, and the mouse-gut microbiome analysis illustrates how LODI separates persistent subject-feature abundance trajectories from shared variation across OTUs.

Several extensions could increase flexibility while preserving this separation. The current model uses one loading matrix $\mathbf{\Lambda}$ throughout follow-up, which pools information across time and makes the factor directions directly comparable. A dynamic loading process $\mathbf{\Lambda}(t)$ could permit the OTU-specific dependence pattern itself to evolve, rather than allowing temporal covariance changes only through the factor process and residual variance. Because an unrestricted $J \times K$ loading matrix at every time point would be weakly estimated when $N \ll J$, a practical version would require temporal smoothing, shrinkage toward a common loading matrix, and identifying constraints that remain coherent over time.

The baseline distribution could also vary across biological groups or covariates. A hierarchical Dirichlet process could define genotype- or OTU-specific mixing distributions that share atoms but use different weights (Teh et al., 2006). Alternatively, a dependent Dirichlet process could allow the entire baseline distribution to change flexibly with genotype or other subject-level covariates (Quintana et al., 2022). These extensions may reveal group-specific baseline heterogeneity, although reliable uncertainty quantification would require more subjects within each group.

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